Cobre · Transaction Corridor Analysis

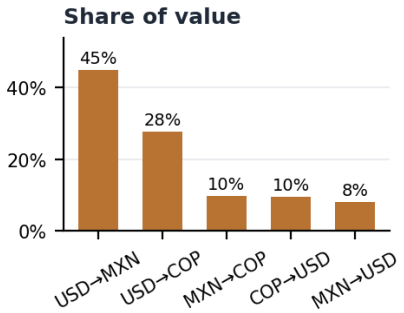
L2 Business Architect Assessment · Candidate submission · Data: 50,000 transactions and 5,000 users, Jul–Dec 2025

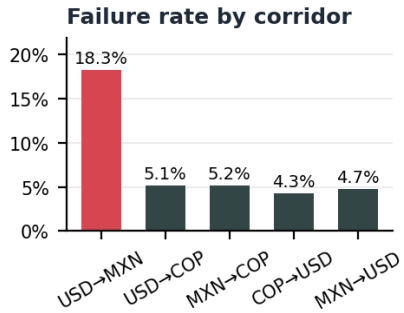
One-page summary

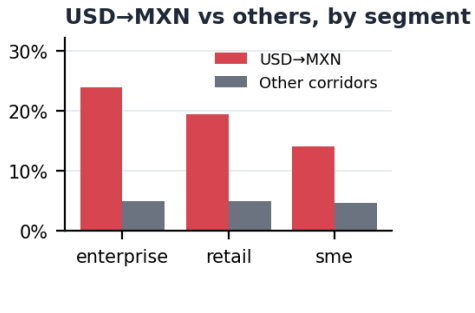
Transactions	Value processed	Success rate	USD→MXN failure	Recoverable / month
50,000	US$281.5M	90.4%	18.3%	US$500K

Key findings

- **USD→MXN is the largest corridor** (35% of transactions, 45% of value) and the only one growing.
- **But it fails 18.3%** vs 4.9% in the other corridors, and concentrates 66% of all Cobre failures.
- **The failure belongs to the corridor:** every segment and amount fails 3-4× more, and it gets worse for payments **above US$10K**. The day of the week has no effect.
- **Mexican and Colombian users behave the same**, and total volume is flat (~273 tx/day for six months).
- **Bringing failures down to 5% is worth US$500K/month.** 2026 recommendation: **prioritize USD→MXN**.







Recommendation: prioritize USD→MXN in 2026

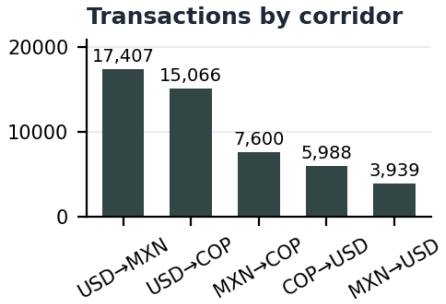
Fix first, then grow. Its success rate is only 81.7%. Bringing failures to 5% unlocks ~US$500K/month without acquiring a single new customer; no other corridor has more than US$2K/month to recover.

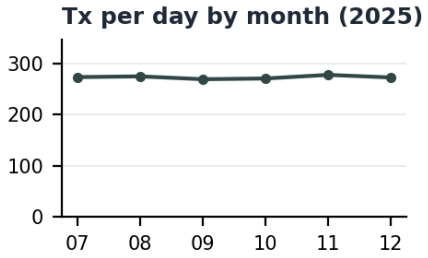
- **Q1 · Stabilize:** Daily dashboard of rejections by code, segment and amount; CLABE and name validation at checkout (retail).
- **Q1–Q2 · Unblock large payments:** KYB/AML pre-approval and agreed limits with the payout bank for payments >US$10K; automatic splitting where allowed.
- **Q2 · Redundancy:** Second MXN payout provider with smart routing based on success rate.
- **H2 · Grow:** Once success is ≥95%, commercial push to US enterprise and SME exporters.

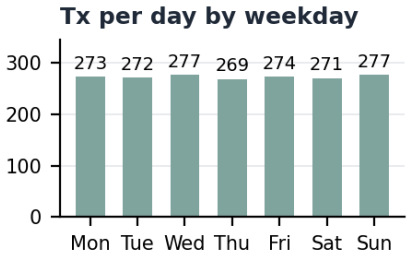
Expected impact: Failure rate from 18.3% to 5%: +US$500K/month (~US$6.0M/year) in recovered revenue, plus retention of enterprise clients who today see 1 in 4 payments fail.

Part 1 · Data analysis

Every answer below is computed in Python; tables and charts are in Cobre_Dashboard.xlsx and Part1_Data_Analysis.xlsx.







Question	Answer
1.1 Corridor performance	
1.1a What are the top 5 payment corridors by transaction volume?	1. USD→MXN (34.8%) · 2. USD→COP (30.1%) · 3. MXN→COP (15.2%) · 4. COP→USD (12.0%) · 5. MXN→USD (7.9%). There are only 5 corridors, so the top 5 is the full set; the first two account for 65% of transactions.
1.1b What are the top 5 corridors by transaction value (total USD)?	1. USD→MXN (US$126.6M) · 2. USD→COP (US$77.8M) · 3. MXN→COP (US$27.5M) · 4. COP→USD (US$26.9M) · 5. MXN→USD (US$22.6M). USD→MXN concentrates **45.0% of value** with 34.8% of transactions.
1.1c Which corridors have the highest success rates?	COP→USD leads with 95.7%. All corridors are around 95% except **USD→MXN, which drops to 81.7%**: it is the only reliability problem.
1.1d What is the average transaction size by corridor?	USD→MXN US$7,271 · MXN→USD US$5,750 · USD→COP US$5,165 · COP→USD US$4,495 · MXN→COP US$3,621. The mean is ~1.8× the median in every corridor: a few large enterprise payments pull the average up.
1.2 User behavior	
1.2a What's the average number of transactions per user?	**10.0 transactions per user** (median 10, max 21). All 5,000 registered users transacted at least once.
1.2b Who are the top 10 users by transaction volume?	The leader is USR_4714 with 21 transactions; the top 10 make between 20 and 21, only 2× the average. No user concentrates volume.
1.2c How does user behavior differ between Mexico and Colombia?	Mexico accounts for 59.1% of transactions because it has more users, but **per user they behave the same**: 10.0 vs 10.0 tx/user, ticket US$5,595 vs US$5,679, success 90.4% vs 90.5%, same corridor and segment mix. The real difference is the destination: money **arriving** in Mexico fails 18.3% vs 5.1% for money arriving in Colombia.
1.3 Time patterns	
1.3a What days of the week have the highest transaction volume?	Sunday has the highest daily average and Thursday the lowest, but the gap between them is only 2.8%: **there is no dominant day**; volume is even across the week.
1.3b Are there any notable hourly patterns?	All activity happens between 8:00 and 19:59 (banking hours), with no overnight transactions. The peak is at 10:00, but the variation across hours is 4.8%: within the window volume is flat.
1.3c Has transaction volume been growing or declining over the 6-month period?	**It is flat.** 274 tx/day in July vs 273 in December (-0.3%); the linear trend is 0.08% per month. Only USD→MXN is growing in value (+4.2% Q4 vs Q3).

Part 2 · Why does USD→MXN fail? Root cause and impact

Write-up · 267 words

Diagnosis. USD→MXN fails 18.3% of transactions (19.8% of value), 3.7× the other corridors (4.9%), and drives 66% of all Cobre failures. The data shows a higher rate than the 15% stated in the brief.

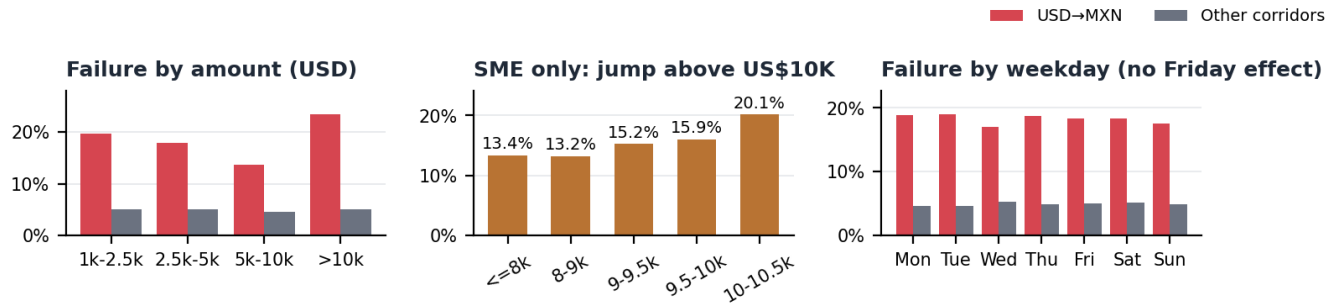
Patterns. (1) Every segment fails more than in other corridors, with a gradient: enterprise 23.9%, retail 19.5% and SME 14.1%; retail drives 52% of failures by volume. (2) Amount matters: payments >US$10K fail 23.4% and, within SME, the rate jumps from 13.7% to 20.1% when crossing US$10K. (3) The day does **not** matter: Fridays fail 18.4% and the weekly range is 17.0%–18.9% (chi², p=0.53); neither do hour or month. Failures are spread across 2,341 users: this is not a single customer.

Three possible root causes. H1: a compliance/AML rule or payout-bank limit on payments >US$10K. H2: invalid beneficiary data (CLABE, name) in retail. H3: a structural issue with the MXN provider or settlement route that raises the baseline failure rate across the whole corridor.

Validation. Analyze rejection codes by segment and amount; compare failure rates by provider/payout bank; pilot CLABE validation at checkout and pre-approval of payments >US$10K with an A/B test.

Impact. At a 0.5% fee, the sample loses US$21K/month; scaled to Cobre's US$1.5B/month that is **US$669K/month**. Cutting failures to 5% recovers **US$500K/month (US$6.0M/year)**; under the brief's assumptions (35%, 15%) it would be US$262K/month.

Recommendation. Attack H1 and H3 first, since they concentrate the lost value: agree limits and pre-approval for large payments with the payout bank, and test a second MXN provider. In parallel, add CLABE validation for retail (H2) and track the failure rate weekly by segment and amount band.



Revenue impact (fee 0.5%; gain = (failed value % − 5%) × corridor value × fee)

Scenario	Monthly value	Failed value %	Revenue lost / month	Gain at 5% / month	Annual gain
Sample (data as is)	US$21,094,515	19.8%	US$20,921	US$15,647	US$187,768
Scaled to US$1.5B/month	US$674,500,531	19.8%	US$668,952	US$500,327	US$6,003,927
Brief scenario (35%, 15%)	US$525,000,000	15.0%	US$393,750	US$262,500	US$3,150,000

Part 3 · Strategic recommendation memo

256 words

To: Management · From: Business Analyst · Subject: Corridor to prioritize in 2026

Prioritize USD→MXN.

Why this corridor

- **Scale:** It moves 45% of processed value; scaled to Cobre's volume it generates ~US$2.7M/month in fees, 1.4× USD→COP.
- **Growth:** It is the only large corridor that is growing: +4.2% in daily value in Q4 vs Q3, while USD→COP fell 1.3%.
- **Hidden upside:** Its success rate is only 81.7%. Bringing failures to 5% unlocks ~US$500K/month without acquiring a single new customer; no other corridor has more than US$2K/month to recover.
- **Market:** US→Mexico is the world's largest remittance corridor and the region's main B2B flow, with mature payment rails (SPEI).

Decision matrix: Weighting revenue (30%), growth (15%), reliability (15%), recoverable revenue (20%), market (10%), simplicity (10%), USD→MXN scores 0.85 vs 0.53 for USD→COP. The result is robust: it still wins when any single criterion is removed.

Recommended actions

- **Q1 · Stabilize:** Daily dashboard of rejections by code, segment and amount; CLABE and name validation at checkout (retail).
- **Q1–Q2 · Unblock large payments:** KYB/AML pre-approval and agreed limits with the payout bank for payments >US$10K; automatic splitting where allowed.
- **Q2 · Redundancy:** Second MXN payout provider with smart routing based on success rate.
- **H2 · Grow:** Once success is ≥95%, commercial push to US enterprise and SME exporters.

Expected impact: Failure rate from 18.3% to 5%: +US$500K/month (~US$6.0M/year) in recovered revenue, plus retention of enterprise clients who today see 1 in 4 payments fail.

Risks: Root causes remain hypotheses until rejection codes are reviewed; market and simplicity scores are qualitative judgments.

Decision matrix (min-max normalized, weighted)

Corridor	% of value	Monthly revenue (scaled)	Growth Q4 vs Q3	Success rate	Recoverable / month	Market (1-5)	Simplicity (1-5)	Score
USD_MXN	45.0%	US$2,703,550	4.2%	81.7%	US$500,327	5	4	0.85
USD_COP	27.6%	US$1,968,009	-1.3%	94.9%	US$1,871	4	4	0.53
MXN_USD	8.0%	US$572,738	1.4%	95.3%	US$562	3	3	0.31
COP_USD	9.6%	US$687,278	1.3%	95.7%	US$0	2	2	0.25
MXN_COP	9.8%	US$695,812	-2.4%	94.8%	US$824	3	2	0.19

Appendix · Use of AI, data quality and deliverables

Use of AI tools

- Claude (Anthropic) was used to refactor the initial scripts into a modular package with a single metrics function (summarize) instead of five duplicated groupby blocks, to suggest the root-cause cuts and the chi² test to rule out the day-of-week effect, and to draft the write-up and memo.
- Own validation: the patterns stated in the brief were checked against the data. Two do not hold: Friday failures (18%) match every other day, and the segment with the highest failure rate is enterprise, not retail (retail does drive more failures by volume). A US$10K threshold test within SME was added to separate the amount effect from the segment effect.
- Python is the single source of truth: every figure and text is generated by python main.py; the Excel workbook and this document only load and display those results.

Data quality and assumptions

- Clean data: no empty cells, no duplicates, no orphan users, and corridors consistent with origin/destination.
- The segment in transactions.csv differs from users.csv in 61% of rows and 4,970 users appear with more than one segment. The transaction segment is used (it is the one the brief describes) and the mismatch is flagged as a data-governance task.
- Data ends on 2025-12-30: monthly metrics are computed per day with data.
- Assumptions: 0.5% fee on successful payments; Cobre's real volume of US$1.5B/month with the same mix as the sample; target failure rate 5%.
- Qualitative scores (Part 3): market 5 = largest remittance corridor (USD→MXN); simplicity 4 = mature SPEI/ACH rails; 2 = intra-LatAm double conversion or FX outflows subject to exchange controls.

Deliverables

File	Content
Cobre_Assessment_Deliverable.docx / .pdf	This document: summary, write-up, memo, AI note.
Cobre_Dashboard.xlsx	Dashboard and one sheet per part: each question, answer, table and chart.
Part1/2/3 workbooks	Every analysis table, one sheet per table (data quality checks in Part 1).
python/ (code)	Single source of truth. Run: python main.py → build_excel.py → build_document.py.